Ritik Mishra

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Summary

B.Tech in Information Technology graduate focused on Generative AI and Agentic AI, with hands-on experience building LLM-powered applications using Python, LangChain, LangGraph, RAG, MCP, and LLM APIs. Built end-to-end AI systems including an AI Coding Agent, Blog Writing Agent, and RAG Knowledge Base Assistant, with experience in agent orchestration, tool calling, multi-provider LLM fallback, retrieval pipelines, structured generation, and production-oriented application development.

Technical Skills

Programming: Python, SQL
Generative AI & Agentic AI: LLMs, LLM APIs, LangChain, LangGraph, RAG, MCP, Tool Calling, Prompt Engineering, Agent Orchestration, Structured Generation
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, Regression, Classification, Feature Engineering, Model Evaluation
Vector Search & NLP: FAISS, Embeddings, Semantic Search, Hugging Face, NLTK
Backend & Tools: FastAPI, Streamlit, Pydantic, SQLite, REST APIs, Git, GitHub

Projects

AI Coding Agent GitHub

Python, LangGraph, LangChain, MCP, FastAPI, Streamlit

- Built an agentic coding system using LangGraph and MCP that plans coding tasks, inspects project files, modifies code, executes allow-listed commands, and verifies results.
- Implemented dynamic MCP tool discovery, human-in-the-loop approval, workspace protection, verification/retry workflows, and multi-provider LLM fallback for resilient execution across coding tasks.
- Added task state management and controlled command execution to safely coordinate code changes, validation, and execution within a local workspace.

Blog Writing Agent GitHub — Live Demo

Python, LangGraph, LangChain, Tavily, Streamlit

- Built an AI-powered blog writing agent using LangGraph and LangChain with topic routing, web research, planning, content generation, and final reduction workflows.
- Integrated Tavily for web research, structured outputs for planning, and parallel content generation with Streamlit for interactive blog creation, refinement, and export.

RAG Knowledge Base Assistant GitHub — Live Demo

Python, LangChain, LangGraph, FAISS, Streamlit

- Built a PDF-based Retrieval-Augmented Generation assistant using LangChain, LangGraph, FAISS, embeddings, and LLMs for context-aware question answering.
- Implemented document ingestion, chunking, semantic retrieval, conversational state management, and real-time streaming responses through an interactive Streamlit interface.

Certifications

Generative AI: Introduction to Generative AI Studio – Simplilearn SkillUp; Generative AI for Everyone – Simplilearn SkillUp
Other: SQL (Basic) – HackerRank; Introduction to Machine Learning – Kaggle; Python Coder – Kaggle; Data Science 101 – IBM

Education

Gautam Buddha University Greater Noida, India
B.Tech in Information Technology *2022–2026 — CGPA: 8.35*